Instruction to Generate SPA Figures using Matlab code

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Instructions for using the Matlab codes to generate the SPA figures

1. Place all the provided Matlab codes and the .mlab file (output of transition_dipole.py) in the same directory.
   - The .mlab file is also an output of the Fortran code, but so is the spectrum_spa which is the same as what this code generates
     - See the Fortran section of the Nondegenerate_ETPA_Figure for more info on Fortran code
     - The benefit to using this code instead of the fortran is that this code can generate SPA data for multiple models at a single time
   - This code will generate multiple SPA spectra data if more than 1 .mlab file is provided
2. In the absorption_lorentzian.m code
   - Set line 2 to based on desired type of spectra
     - 1 for absorption spectra
     - 2 for transient absorption spectra
   - Set line 3-7 with desired values
     - Energies should be given in eV
3. Run the absorption_lorentzian.m code
4. Will output `spectrum_<.mlab name>` files (e.g. spectrum_PBE for PBE.mlab)
   - This is the SPA spectrum data (First column is energy values in eV, second column is absorption intensity)
5. Plot the spectrum with your desired software. I have provided a sample matlab code that I used to plot the SPA when doing the functional comparisons.
   - absorption_functional.m